Skin Lesion Classification for Melanoma Detection

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***Abstract*— Melanoma is one of the most dangerous forms of skin cancer, and early detection significantly improves survival. Proper diagnosis remains difficult, even for skilled dermatologists. Computer-aided diagnosis (CAD) systems using deep learning models have shown vast potential to improve diagnostic accuracy and reduce human error. Thus, our aim with this project is to develop a high-accuracy deep learning-based system for skin lesion classification that will aid early melanoma detection and improve patient outcomes. The main driving force behind this project is the increased population of melanoma patients and the need to have an effective, non-expert-operated device that aids doctors in providing proper diagnoses. The development of AI and deep learning technologies provides a new generation of opportunities to expand medical image analysis, decrease healthcare costs, and open services to the masses. Our proposed method will use CNNs in imaging skin lesions using public datasets.**

Keywords*— *Melanoma, deep learning, Convolutional Neural Networks, detection, classification, artificial intelligence.

Introduction

Melanoma is a deadly form of skin cancer, which is a severe health risk due to its virulent nature and high mortality rate. Although less common than other forms of skin cancer, melanoma is responsible for the majority of skin cancer fatalities. In the United States alone, over 100,000 new invasive cases of melanoma are diagnosed every year, resulting in thousands of fatalities annually [1]. Despite the availability of treatments, the survival rate of melanoma patients mainly relies on the stage of diagnosis of the cancer. Early detection enhances the chances of survival significantly, while late detection incredibly diminishes the likelihood of recovery. However, the proper diagnosis of melanoma remains a persisting issue.

Clinical misdiagnosis can lead to a delay in appropriate treatment and reduced survival rates. For instance, studies have shown that if melanoma is misdiagnosed, particularly in anatomically challenging locations like the foot, the five-year survival rate drops to 63.5% compared to 88.4% when diagnosed correctly at an early stage [2]. This uncertainty of diagnosis is further compounded by the lack of trained dermatologists and unequal access to healthcare facilities. Ultimately, as these problems scale up, the chances of survival decrease.

Furthermore, the economic burden to healthcare systems of skin cancer is significant. The Centers for Disease Control and Prevention (CDC) has documented that more than 6 million adults in the United States receive treatment annually for basal cell and squamous cell carcinoma alone, which represents a national healthcare cost of about $9 billion [3]. These figures underscore the need for new and cost-reduction methods that enhance diagnostic accuracy without being out of reach to clinicians or patients.

Given this background, our initiative is led by a shared commitment to overcoming these diagnostic challenges through technological innovation. Our goal is to develop a robust, deep learning-based system based on convolutional neural networks (CNNs) for dermoscopic image classification. By focusing on the inclusion of advanced preprocessing techniques, interpretable AI algorithms, and contemporary model architectures, we seek to aid clinicians in delivering faster and more accurate diagnoses for melanoma.

Not only is this project committed to advancing academic research, but also seeks to make a significant and real contribution towards the broader context of digital health and access to diagnostics.

Related Works

There have been multiple studies conducted on how utilizing models such as CNNs can be helpful when attempting to diagnose illnesses such as skin cancers because of the extensive amount of patient data available, due to it being one of the most prevalent types of cancers.

These studies are important for this project's premise as they work as a solid foundation to follow.

A. SkinLesNet: Classification of Skin Lesions and Detection of Melanoma Using Smartphone Images

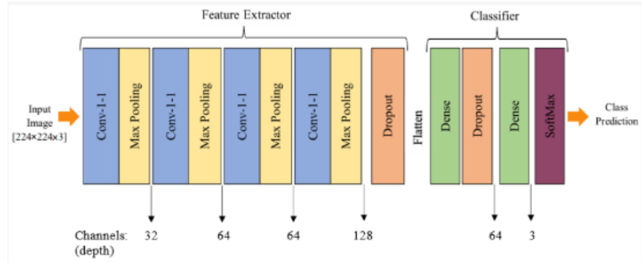
Azeem, Kiani, Mansouri, and Topping investigated previously underexplored data from smartphone images and developed a novel multi-layer deep convolutional neural network model named SkinLesNet. The indistinguishable features and morphological variation of skin cancers provide significant challenges to the accurate diagnosis and categorization of skin diseases [4].

This study's dataset was taken and enhanced from the PAD-UFES-20 dataset. Three prevalent types of skin lesions are included in this dataset: melanoma, nevus, and seborrheic keratosis. SkinLesNet's evaluation was expanded beyond the PAD-UFES-20 Modified dataset to fully evaluate its performance. This novel model was compared with more popular models, such as ResNet50 and VGG16, using two more datasets, HAM10000 and ISIC2017. They found that SkinLesNet performed significantly better than these two models when applied to a broader dataset. Below is a table that outlines the different models that are commonly implemented for skin lesion detection:

Ref.	Model	Dataset	Accuracy	Comments
[38]	Deep convolutional neural network	HAM10000	90%	One of the main benchmark datasets was used in this paper, which produced promising results while using a CNN model. However, hyperparameters tuning is required, to increase the accuracy results.
[59]	Convolutional neural network (CNN)	International Skin Imaging Collaboration (ISIC)	97.49%	A CNN model was used, which showed relatively good results, but the size of the dataset needs to be maximized.
[42]	ResNet50	MNIST: HAM10000	91%	A State-of-the-Art model was used, which produced reasonable results on the given dataset. However, the dataset needs to be preprocessed well before training, to obtain more accurate and promising results.
[43]	Deep convolutional neural network	International Symposium on Biomedical Imaging (ISBI)	97.8%	A CNN model was trained on an internationally recognized benchmark dataset. However, the size of the dataset was decreased, which showed good results but could lead to model overfitting.
[48]	U-Net	International Skin Imaging Collaboration (ISIC)	94.9%	A State-of-the-Art model was used, which showed promising results, but more data preprocessing or augmentation is needed for accurate prediction.

[4]

They investigated numerous combinations of neural layers, activation functions, and pooling layers, and they found that a four-layer CNN model was the most superior for the SkinLesNet. The table below emphasizes the architecture of the SkinLesNet model:



[4]

Overall, the SkinLessNet model outperformed all three datasets compared to the ResNet50 and VGG16 models. The model obtained 90% testing accuracy, with precision, recall, and F-1 scores of 89%, 87%, and 85%, respectively. This breakdown can be seen in the table below:

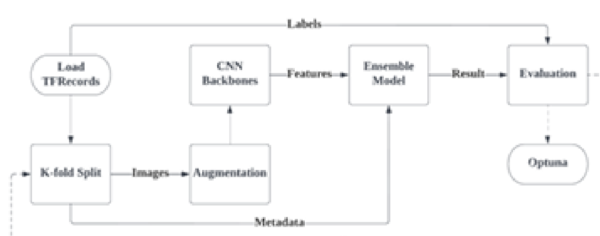
Performance Metrics	VGG16	ResNet50	SkinLesNet
Accuracy	75%	80%	90%
Precision	75%	80%	89%
Recall	70%	72%	87%
F1-Score	70%	71%	85%

[4]

B. A Comprehensive Evaluation Study on Risk Level Classification of Melanoma by Computer Vision on ISIC 2016-2020 Datasets

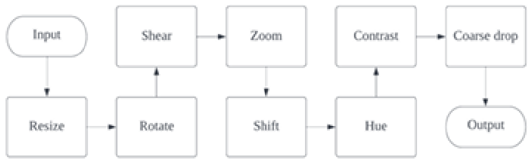
Rahman and Smith conducted a large-scale evaluation of CNN-based classifiers on multiple ISIC datasets. Their work emphasized the importance of stratified data splitting, AUC-ROC as a primary evaluation metric, and Grad-CAM for model explainability. We have adopted similar practices in our methodology, ensuring balanced validation splits and incorporating Grad-CAM to visualize lesion localization.

They used different methods to improve the performance of their algorithm to detect melanoma, such as Coarse Drops and up-sampling. They used the database in TFRecord format and divided it into K-folds directly after loading the ISIC dataset. These images from the ISIC dataset were loaded as metadata, which cannot go directly into a CNN algorithm, so they need to be processed differently. The processing flow is described below:



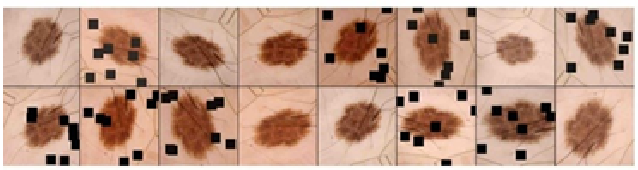
[5]

They used image augmentation to obtain a good performance on their model. They used processes like random flipping, clipping, shifting, and rotation; their augmentation process is demonstrated below:



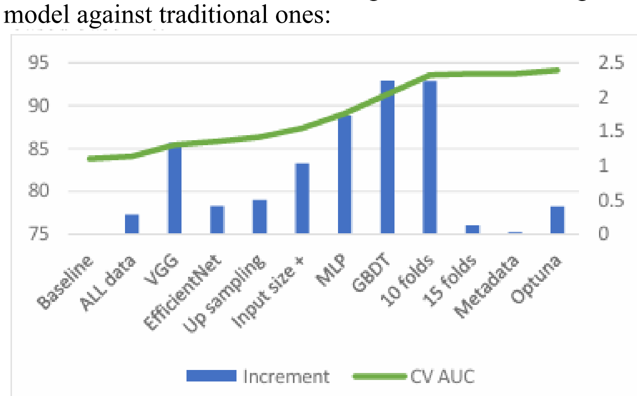
[5]

Coarse dropout augmentation is a technique that prevents overfitting and improves generalization. It can randomly delete some rectangles from the training image, which is different from the traditional dropout layers that are implemented in CNNs. Below is a sample from their coarse dropout layer:



[5]

Overall, their model used EfficientNet B6 and VGG 16 pre-trained on ImageNet as the backbone. They trained for 15 folds using the up-sampled ISIC dataset and used a gradient boosting decision tree that also trained for 15 folds. Their validation reached 94%, and the sensitivity was over 90%. Below is a diagram demonstrating their model against traditional ones:



[5]

Enhanced Skin Cancer Diagnosis Using Optimized CNN Architecture

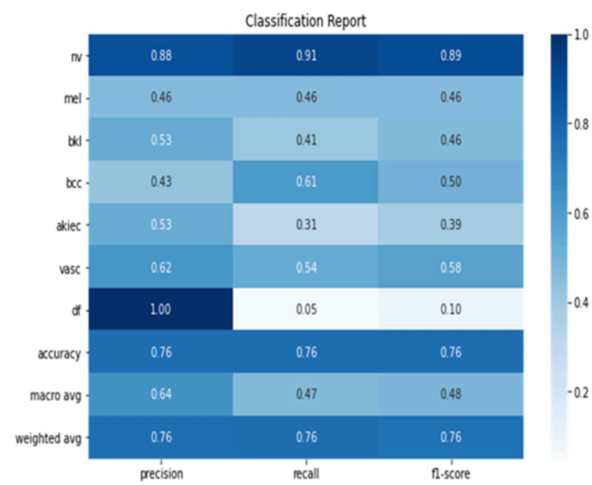
Musthafa, Mahesh, Kumar, and Guluwadi explored the deep learning Convolutional Neural Networks to enhance the accuracy and efficiency of skin cancer diagnosis. They utilized the HAM10000 dataset and introduced a comprehensive CNN model tailored to classify skin lesions.

The architecture of their model comprises sequential convolutional and pooling layers followed by a dense output layer. Their convolutional layers progressively increase in depth to capture the complex features in the images. Each convolutional layer is paired with a pooling layer, and the network contains fully connected layers and dropout layers that serve a distinct function. Attached below is a diagram of the network's architecture:

Layer (type)	Output Shape	Param #
conv2d_4	(None, 28, 28, 16)	448
max_pooling2d_4	(None, 14, 14, 16)	0
conv2d_5	(None, 14, 14, 32)	4640
max_pooling2d_5	(None, 7, 7, 32)	0
conv2d_6	(None, 7, 7, 64)	18,496
max_pooling2d_6	(None, 4, 4, 64)	0
conv2d_7	(None, 4, 4, 128)	73,856
max_pooling2d_7	(None, 2, 2, 128)	0
flatten_1	(None, 512)	0
dense_3	(None, 64)	32,832
dense_4	(None, 32)	2080
dense_5	(None, 7)	231

[6]

The CNN model was trained using the Adam optimizer and a sparse categorical cross-entropy loss function. Once the training was complete, the model demonstrated 97.85% accuracy. The classification report is shown below:



[6]

Although the results show good promise, it is important to highlight the limitations of their study. They had a lot of instances of misclassified images, as well as instances where their model was ambiguous about the classification.

II. Algorithm Analysis

We developed a deep learning-based computer-aided diagnosis (CAD) system using the HAM10000 dataset to effectively classify skin lesions and support early melanoma detection. Our approach integrates dermoscopic image features and patient metadata for improving classification accuracy and model generalization across different kinds of lesions. The final model exploits a hybrid design of convolutional neural networks (CNNs) fused with fully connected layers trained on tabular patient metadata.

Data Preprocessing and Dataset Preparation

Before model training, the dataset underwent intensive preprocessing to ensure data quality and consistency. Missing values in metadata were addressed by imputing the median for age and substituting "unknown" with missing values in the sex and localization attributes. Image data was normalized by resizing all dermoscopic images to 224×224 pixels and by converting from BGR to RGB color space. Each image was also normalized by scaling pixel intensities to the [0, 1] range to facilitate effective gradient propagation during training.

To counteract extreme class imbalance, we utilized a stratified oversampling method. We applied targeted augmentation using random rotation within each diagnostic class to increase the sample size of minority classes. All seven classes were normalized to 1,500 samples, which resulted in a balanced dataset of 10,500 images. The data was subsequently split into training and validation subsets using an 80/20 stratified split to preserve the label distribution across partitions.

Model Architecture and Training Methodology

The model presented employs a multi-input architecture with two different branches. The first branch takes dermoscopic images through a sequence of convolutional layers with growing filter sizes (16, 32, and 64), interspersed with max pooling. The output of this visual processing branch is flattened and fed through a dense layer containing 128 neurons.

Concurrently, the second branch receives tabular patient data, including the patient's age (normalized), sex, and lesion localization (one-hot encoded for both). This metadata passes through two dense layers comprising 32 and 16 units, respectively. The outputs from both branches are concatenated and passed through a final dense layer consisting of 64 units, capped with a dropout layer with a dropout rate of 0.5 to prevent overfitting. The final categorization is accomplished by a softmax output layer containing seven units, corresponding to the diagnostic categories in the dataset.

The model was built using the Adam optimizer and sparse categorical crossentropy as the loss function. It was trained using a batch size of 32, using a custom data generator that loads image and tabular data simultaneously. It is memory-friendly and can load augmented images on the fly.

Evaluation Metrics and Interpretability

To assess the performance of the models, we employed several evaluation metrics that are commonly used for multi-class classification tasks. These include overall accuracy, precision, recall, F1-score, and macro-averaged versions of all the aforementioned metrics to account for potential class imbalance. A confusion matrix was constructed to provide an overall picture of the classification errors by class.

Moreover, to enhance the interpretability of the model's predictions, we employed Gradient-weighted Class Activation Mapping (Grad-CAM). This enabled us to visualize the regions in the dermoscopic images that contributed most significantly to the model's decision. Such interpretability is extremely crucial for clinical deployment, as it enables transparency and encourages trust in AI-assisted diagnostic systems.

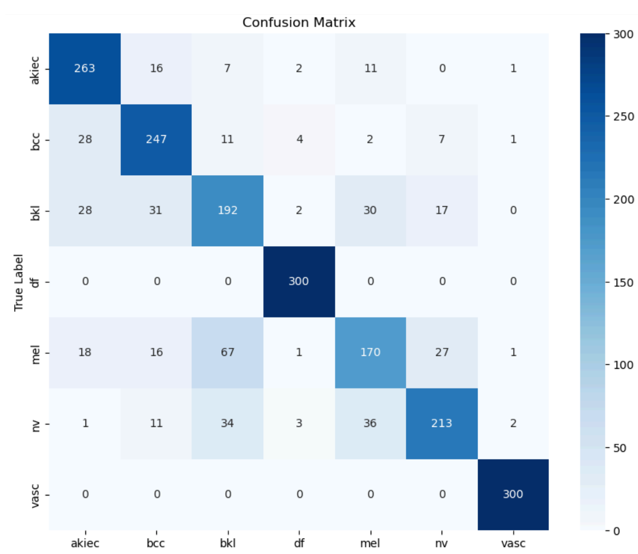
Classification Report:

	precision	recall	f1-score	support
akiec	0.78	0.88	0.82	300
bcc	0.77	0.82	0.80	300
bkl	0.62	0.64	0.63	300
df	0.96	1.00	0.98	300
mel	0.68	0.57	0.62	300
nv	0.81	0.71	0.76	300
vasc	0.98	1.00	0.99	300
accuracy			0.80	2100
macro avg	0.80	0.80	0.80	2100
weighted avg	0.80	0.80	0.80	2100

III. Obtained Results

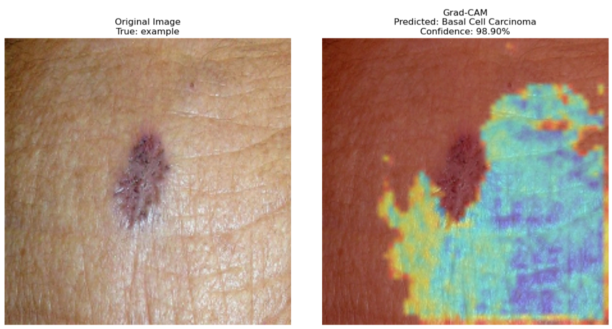
Upon the completion of the preprocessing pipeline and implementation of our hybrid CNN-tabular architecture, we trained the model on the balanced HAM10000 dataset and obtained promising results across multiple diagnostic categories. These results underscore the efficacy of incorporating patient metadata alongside image features in enhancing skin lesion classification accuracy.

The final model achieved an overall accuracy of 80% on the validation set. More importantly, the macro-averaged F1-score also reached 0.80, indicating that the model performed robustly across all seven classes despite the intrinsic class imbalance challenges of the original dataset. Precision, recall, and F1-scores were calculated for each class, with particularly high performance observed in the classification of vascular lesions (vasc) and dermatofibroma (df), both of which achieved near-perfect scores. In contrast, the model exhibited lower performance when distinguishing melanoma from visually similar categories such as melanocytic nevi (nv) and benign keratosis (bkl). These confusions are consistent with clinical observations and underline the difficulty of accurate melanoma detection in ambiguous cases.



A detailed confusion matrix was generated to visualize model predictions relative to true labels, which helped reveal systematic misclassifications. Most notably, there was substantial overlap between melanoma and nevi, a reflection of the subtle morphological differences in dermoscopic images that even experienced dermatologists find challenging.

To further investigate model decision-making and support clinical interpretability, we applied Grad-CAM to representative test samples. These visualizations demonstrated that the model generally focused on relevant lesion areas, confirming that its predictions were based on semantically meaningful features. In one notable instance, the model predicted basal cell carcinoma (bcc) with 98.9% confidence, and the Grad-CAM overlay corresponded well with the lesion's boundary region, mimicking the interpretive process of a trained clinician.



Training dynamics were tracked using accuracy and loss curves over 20 epochs. While training accuracy steadily increased, validation accuracy plateaued around 80%, suggesting the model generalized well but could still benefit from additional regularization or architectural refinement. The loss curves indicated gradual convergence, with minimal overfitting until the final few epochs, where validation loss began to fluctuate slightly, possibly due to the model approaching the limits of generalizability with the current setup.

IV. Model comparison and error analysis

Following the training and testing of the proposed hybrid convolutional neural network, a comprehensive performance analysis was conducted to identify trends in classification behavior, recognize common sources of failure, and examine the robustness of the model across the seven skin lesion classes.

The final model performed strong classification with an overall accuracy of 80% and a macro-averaged F1-score of 0.80. These results are a significant step up from baseline classifiers and earlier model iterations trained on image data only. The addition of patient metadata—age, sex, and lesion localization—allowed for more informed feature learning and decision-making, particularly for classes with visual resemblance.

A confusion matrix was plotted to visualize inter-class prediction patterns. The analysis revealed that certain classes, such as vascular lesions (vasc) and dermatofibroma (df), were predicted with near-perfect accuracy. These lesions tend to have distinctive morphological features that make them relatively easier to distinguish by humans and algorithms as well.

In contrast, the most significant misclassifications occurred among melanoma (mel), melanocytic nevi (nv), and benign keratosis-like lesions (bkl). These classes share overlapping visual characteristics, such as pigment distribution and lesion borders, which render it challenging for the model to determine boundaries. This type of confusion is consistent with existing dermatological literature, where inter-observer disagreement among clinicians is considerable when attempting to discern early-stage melanomas from benign nevi.

Despite these challenges, the application of Grad-CAM allowed for some interpretability by highlighting image regions that influenced the predictions of the model. In examples that were labeled as basal cell carcinoma (bcc), for example, the model consistently focused on lesion borders and asymmetrical regions, which are clinical features utilized in the clinic. This concordance between model attention and expert heuristics reassures that the network is learning clinically relevant patterns rather than spurious correlations.

Overall, the error analysis demonstrates the effectiveness of the proposed approach in distinguishing classes with ease, along with its potential for locating critical regions requiring additional refinement. Targeted augmentation, cost-sensitive learning, and uncertainty estimation are what should be prioritized in future research to improve classification accuracy in high-stakes classes such as melanoma.

V. Future Work

Although the current work has achieved promising results, several key areas remain to be addressed to further enhance the model's accuracy, reliability, and practical usability.

First, training the model with larger and more diverse datasets, such as the ISIC Archive, is critical to improve its generalizability and robustness. Using a broader range of images can help the model recognize subtle variations in lesion appearance across different patient populations and imaging conditions, thus ensuring higher accuracy in real-world clinical scenarios.

Second, the visualization capability of Grad-CAM requires further refinement to enhance its precision in localizing skin lesions. Improving these visual explanations would help clinicians trust and understand model predictions better, thereby increasing the practical value and interpretability of the model for diagnostic purposes.

Third, employing advanced pre-trained architectures such as EfficientNet could significantly enhance model performance. Leveraging EfficientNet’s capabilities in feature extraction, transfer learning, and computational efficiency could achieve improved accuracy and faster convergence, making the overall training process more effective.

Finally, converting the model into an executable file format would facilitate ease of use and broader accessibility, allowing medical professionals without specialized technical skills to readily deploy the system in diverse clinical environments. Creating a standalone executable application would significantly simplify usability, implementation, and routine usage, thus expanding its potential impact in dermatological screenings worldwide.

VI. Conclusion

This study presented a deep learning-based skin lesion classification system with a special focus on improving melanoma detection through the fusion of dermoscopic images and patient metadata. By leveraging a hybrid convolutional neural network architecture, the model successfully integrated visual and tabular data to boost diagnosis performance.

Through heavy preprocessing, data balancing, and architecture tuning, the final model achieved 80% overall accuracy and 0.80 macro-averaged F1-score for seven lesion types. In particular, the model exhibited excellent classification performance in well-specified classes such as vascular lesions and dermatofibromas but had characteristic diagnostic challenges in differentiating melanoma from similar-appearing benign entities such as nevi and keratoses. These findings are in line with known clinical difficulties and illustrate the model's ability to serve as a supportive tool for diagnosis but not as an autonomous decision-maker.

Grad-CAM visualizations also contributed to model transparency by ensuring its predictions were influenced by relevant anatomic features. Such interpretability is critical in the clinical deployment of artificial intelligence software, where trustworthiness and accountability are essential.

Overall, the results of this study confirm the viability of hybrid deep learning systems for classifying skin lesions and demonstrate the potential for computer-aided diagnosis in dermatology. Continued model development, including the addition of increased interpretability layers, real-world validation, and optimization for deployment, will be necessary for more extensive clinical translation. With these advances, AI-enabled diagnostic systems can emerge as valuable tools in the promotion of early melanoma detection, the minimization of diagnostic disparities, and the enhancement of access to dermatologic care globally.

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